23.05.2025 (M)

SOMAIYA VIDYAVIHAR UNIVERSITY

Semester: Jan2025 –April 2025		
Maximum Marks: 100 Examination: ESE Examination		Duration:3 Hrs.
Programme code: 41 Programme: Minor Programme in Artificial Intelligence and Machine Learning	Class: LY	Semester: VII (SVU 2020)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: COMP	
Course Code: 116m41C701	Name of the Course: Deep Learning	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	Explain the significance of 'Momentum' in deep networks.	5
ii)	What is the difference between shallow and deep neural networks? Give examples.	5
iii)	Explain the importance of regularization in deep learning.	5
iv)	What are autoencoders? State any two applications.	5
v)	What is vanishing gradient problem	5
vi)	Explain non-convex optimization for deep networks.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	What is the difference between Conv2D and Pooling layers in CNNs? Explain with examples.	5
ii)	Describe the image classification process using CNN.	5
	OR	
Q2 A	Compare and contrast CNN and RNN with architecture diagrams and use cases.	10
Q 2 B		**10**
i)	Explain the architecture of AlexNet convolutional neural network	10
ii)	Explain the working of ResNet with a block diagram and its skip connection advantage.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Explain vanilla autoencoder with the help of architecture diagram.	10
ii)	With a neat sketch, explain the architecture of LSTM. How does it solve the vanishing gradient problem?	10
iii)	What are Generative Adversarial Networks (GANs)? Explain generator and discriminator interaction.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Explain the working of Gated Recurrent Unit.	10
ii)	You are given the architecture of a simple CNN model: Input: 32×32×3 RGB image Layer 1: Conv2D - Filters: 16 - Kernel size: 3×3 - Stride: 1 - Padding: 'valid' Layer 2: MaxPooling2D - Pool size: 2×2 - Stride: 2 Layer 3: Conv2D - Filters: 32 - Kernel size: 3×3 - Stride: 1 - Padding: 'same' Layer 4: Flatten Layer 5: Dense - Units: 64 - Activation: ReLU Layer 6: Dense - Units: 10 (output classes) - Activation: Softmax 1.Calculate the output shape after each layer. 2. Calculate the number of trainable parameters in each trainable layer.	10
iii)	Differentiate between autoencoders and PCA for dimensionality reduction.	10

Que. No.	Question	Max. Marks
Q5	Solve any **four**	**20**
i)	Compare and contrast LSTM and gated recurrent units	5
ii)	What is dropout in neural networks?	5
iii)	Compare stochastic gradient descent (SGD) with batch gradient descent.	5
iv)	Explain the effect of learning rate on model training with a diagram or illustration.	5
v)	Give an application of LSTM for text classification and explain its suitability.	5
vi)	What is overfitting in deep learning models? How can it be mitigated?	5